

Structural Equations of a Two-Country Monetary Union with Sovereign Risk and Constrained Financial Intermediaries

A microfounded exposition of the model in `code/global/`

Model documentation

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Abstract

This document states the full structural environment of the two-country heterogeneous-agent model of a monetary union solved in `code/global/`. The economy consists of two countries, D (the risky sovereign; Greece) and F (the core; Germany), each populated by incomplete-markets households, competitive producers, capital producers with adjustment costs, monopolistically competitive retailers, and Gertler–Karadi/Bocola financial intermediaries subject to an occasionally-binding incentive-compatibility constraint. Only D’s sovereign debt carries default risk, and that risk is *exogenous and priced*: following Bocola (2016, *JPE*), an increase in the priced one-period default probability depresses long-duration bond prices, inflicts mark-to-market losses on intermediary net worth, tightens the leverage constraint, widens lending spreads and contracts output — without default ever being realised along the baseline path. The document derives each block from primitives, states every first-order condition, gives the closed form of the leverage multiplier, defines the recursive competitive equilibrium and the exact market-clearing system that the projection solver imposes, and documents — explicitly — the approximations that separate the solved object from the primitive environment.

Contents

1	Environment, timing and notation	2
1.1	Overview	2
1.2	Timing convention	3
1.3	Notation	3
2	Households	3
2.1	Preferences and the individual problem	3
2.2	First-order conditions	4
2.3	Distribution and aggregation	5
2.4	The aggregate closure used in the recursive solution	5
3	Production, trade and capital	6
3.1	Final goods and the terms of trade	6
3.2	Retailers and marginal cost	6
3.3	Intermediate producers and the working-capital wedge	7
3.4	Capital producers, Tobin’s q , and the return on capital	7
4	Financial intermediaries	8
4.1	The balance sheet	8
4.2	Gross wealth and the deposit-obligation state	8
4.3	The banker’s problem and the franchise value	9
4.4	The agency friction and the incentive-compatibility constraint	9

4.5	First-order conditions and the closed-form multiplier	9
4.6	The three pricing conditions in explicit form	10
4.7	Net worth accumulation, entry and exit	11
4.8	Steady-state calibration of the agency friction	11
5	Sovereign risk and the default event	12
5.1	Debt instrument	12
5.2	The exogenous, priced risk process	12
5.3	The default event	12
5.4	The expectation operator	13
5.5	Branch stochastic discount factors and the endogenous premium	13
6	Government and policy	13
6.1	The consolidated government budget constraint	13
6.2	The fiscal rule	14
6.3	The monetary block of the union	14
6.4	The TPI/OMT backstop	14
6.5	Macroprudential policy	15
7	Equilibrium	15
7.1	Market clearing	15
7.2	Recursive competitive equilibrium	16
7.3	The solved system, equation by equation	17
7.4	Definitions of the key endogenous variables	17
7.5	Decomposition of the output response	17
8	The steady state	18
9	Numerical solution	18
10	Calibration	19
A	Symbol-to-code map	19
B	Summary of the transmission mechanism	19

1 Environment, timing and notation

1.1 Overview

Time is discrete and infinite, $t = 0, 1, 2, \dots$, and a period is one quarter. The world consists of two countries, indexed $X \in \{D, F\}$, forming a monetary union. Each country produces a distinct traded good. Prices are fully flexible, so all equilibrium objects below are *real*; the “monetary union” is modelled at the level at which it binds real allocations, namely a single union-wide funding market for intermediary liabilities together with real interest parity (Section 6.3).

Each country hosts four types of agent:

1. a continuum of **households** facing uninsurable idiosyncratic labour-income risk and a borrowing constraint, who save exclusively in bank deposits and supply labour with GHH preferences;
2. competitive **intermediate producers** who rent capital and hire labour, and who must pre-finance a fraction of the wage bill with intra-period bank credit (Neumeyer–Perri working capital);

3. **capital producers** subject to Jermann (1998) adjustment costs, and monopolistically competitive **retailers** with flexible prices;
4. a representative **financial intermediary** (“bank”) that holds the economy’s productive capital, both sovereign bonds and the working-capital loan book, funds itself with household deposits, and is subject to an agency friction that generates an occasionally-binding leverage constraint.

Each country also has a **government** issuing long-duration Hatchondo–Martinez perpetuities and levying taxes according to a Bohn-type rule, and the union has a **central bank** whose only instrument is a sovereign-bond backstop of the Transmission Protection Instrument (TPI/OMT) type.

1.2 Timing convention

Timing conventions are load-bearing in this model and are stated once, here.

1. **Capital is predetermined** (Bocola, 2016, eq. 6). The capital stock that produces at t was purchased and priced by banks at $t - 1$. Denote by $K_{X,t}$ the stock *producing* at t ; the bank chooses $K_{X,t+1}$ at t . Impact responses of output therefore run through hours alone.
2. **The deposit rate is predetermined.** The gross rate paid on deposits held from t to $t + 1$, $1 + r_{X,t}$, is set at t . It is the return that appears in the household Euler equation formed at t and in the bank’s funding cost at $t + 1$.
3. **Default is realised at the beginning of a period**, before payoffs are collected: a haircut applies to coupons and to the continuation value of the whole outstanding claim.
4. **Working-capital loans are intra-period:** extended at t , repaid at t with interest, and settled inside the bank’s carried obligation, so they never appear as a separate state.

1.3 Notation

Table 1 fixes symbols. Two conventions deserve emphasis. First, p_t is the terms of trade measured as *units of the D-good per unit of the F-good*, so $p_t \uparrow$ is a real depreciation for D. Second, $P_{X,t}$ denotes the bank’s *gross deposit obligation* carried into t (a state variable, following Bocola’s notation), while $P_{X,t}^c$ denotes the CES *consumption price index*. They are distinct objects.

2 Households

2.1 Preferences and the individual problem

Country X is populated by a unit continuum of infinitely-lived households indexed by i . Preferences are of the Greenwood–Hercowitz–Huffman (GHH) form: the consumption–labour composite

$$x_{it} \equiv c_{it} - v(n_{it}), \quad v(n) = \chi_X \frac{n^{1+1/\nu_X}}{1 + 1/\nu_X}, \quad (2.1)$$

enters a CRRA felicity function, so that household i solves

$$\max_{\{c_{it}, n_{it}, a_{i,t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_X^t \frac{(c_{it} - v(n_{it}))^{1-\sigma_X} - 1}{1 - \sigma_X}, \quad (2.2)$$

Table 1: Notation. Country index $X \in \{D, F\}$ suppressed where unambiguous.

Symbol	Meaning	Symbol	Meaning
$C_{X,t}$	aggregate consumption (basket units)	$n_{X,t}$	bank net worth (end of t)
$N_{X,t}$	aggregate hours	$n_{X,t}^g$	gross bank wealth (start of t)
a_{it}	individual deposit holdings	$\mathcal{A}_{X,t}$	bank total assets (market value)
$A_{X,t}$	aggregate real deposits	$P_{X,t}$	gross deposit obligation (state)
e_{it}	idiosyncratic labour efficiency	$\varphi_{X,t}$	marginal value of bank net worth
$Y_{X,t}$	gross output	$\theta_{X,t}$	bank leverage $\mathcal{A}/n$
$K_{X,t}$	capital <i>producing</i> at t	$\mu_{X,t}$	IC multiplier (rescaled)
$I_{X,t}$	investment	λ	divertable asset share
$Q_{X,t}$	Tobin's q (capital price)	f	banker exit/payout share
$w_{X,t}$	real wage (post-wedge)	ω	entrant endowment share
$r_{X,t}$	deposit rate set at t	$\Lambda_{t,t+1}$	household SDF
$r_{X,t}^k$	realised return on capital	$\Omega_{t,t+1}$	bank's augmented SDF
$r_{X,t}^{wc}$	working-capital lending rate	ζ	working-capital share of wage bill
$L_{X,t}$	working-capital loans	δ_b	bond amortisation rate
p_t	terms of trade (D per F)	B_t	D-sovereign stock (face)
$P_{X,t}^c$	CES consumption price index	q_t^X	sovereign bond price
$IM_{X,t}$	imports	s_t	sovereign-risk factor
$NX_{X,t}$	net exports (own-good units)	π_t^d	priced default probability
$Z_{X,t}$	total factor productivity	d_t	realised default indicator
$T_{X,t}^\tau$	lump-sum taxes	h_t	survival factor $1 - d_t(1 - \varrho)$
$\text{Div}_{X,t}$	dividends to households	ϱ	recovery rate
$\mathcal{S}_t$	aggregate state vector	$\varkappa_{\text{tpi}}$	priced TPI activation prob.

subject to the flow budget constraint, expressed in units of the country's consumption basket,

$$c_{it} + a_{i,t+1} = (1 + r_{X,t}^c) a_{it} + \frac{w_{X,t}}{P_{X,t}^c} e_{it} n_{it} + \frac{\text{Div}_{X,t} - T_{X,t}^\tau}{P_{X,t}^c}, \quad (2.3)$$

and to the no-borrowing constraint

$$a_{i,t+1} \geq \underline{a}_X = 0. \quad (2.4)$$

Here a_{it} are real deposits carried into t , the only asset available to households, and

$$1 + r_{X,t}^c \equiv (1 + r_{X,t-1}) \frac{P_{X,t-1}^c}{P_{X,t}^c} \quad (2.5)$$

is the realised real return on the deposit contract signed at $t - 1$ — the predetermined-rate convention of Section 1.2. Dividends $\text{Div}_{X,t}$ are the consolidated payout of retailers, capital producers and exiting bankers (equation (4.21)); $T_{X,t}^\tau$ are lump-sum taxes.

Idiosyncratic labour efficiency e_{it} follows a stationary AR(1) in logs,

$$\log e_{i,t+1} = \rho_e \log e_{it} + \epsilon_{i,t+1}, \quad \epsilon \sim N(0, \sigma_e^2), \quad (2.6)$$

discretised by the Rouwenhorst (1995) method into an n_e -state Markov chain $(\mathbf{e}, \Pi)$ with grid normalised so that $\sum_e \pi^*(e) \mathbf{e}(e) = 1$, where π^* is the stationary distribution of Π . Idiosyncratic risk is uninsurable: markets are incomplete and the only self-insurance vehicle is the deposit account, subject to (2.4).

2.2 First-order conditions

Let $\eta_{it} \geq 0$ be the multiplier on the borrowing constraint. The intertemporal and intratemporal optimality conditions are

$$x_{it}^{-\sigma_X} = \beta_X (1 + r_{X,t}) \mathbb{E}_t \left[\left(\frac{P_{X,t}^c}{P_{X,t+1}^c} \right) x_{i,t+1}^{-\sigma_X} \right] + \eta_{it}, \quad \eta_{it} (a_{i,t+1} - \underline{a}_X) = 0, \quad (2.7)$$

$$\chi_X n_{it}^{1/\nu_X} = \frac{w_{X,t}}{P_{X,t}^c} e_{it}. \quad (2.8)$$

Equation (2.8) is the defining analytical convenience of GHH preferences: the marginal rate of substitution between consumption and leisure is independent of c_{it} , hence of wealth. Labour supply per efficiency unit is therefore *identical across households*, and aggregation over the wealth distribution is exact. Writing $N_{X,t} \equiv \int e_{it} n_{it} d\Phi_t$ for efficiency-weighted aggregate hours and using $\int e d\Phi = 1$, the aggregate labour-supply schedule is

$$\boxed{\chi_X N_{X,t}^{1/\nu_X} = \frac{w_{X,t}}{P_{X,t}^c}} \quad (2.9)$$

which is the object the equilibrium system imposes (residual 3–4 in Section 7.3). The same property implies that the intertemporal margin depends on the composite x_{it} only, so (2.7) can be solved by the endogenous grid method on x .

► *GHH removes the wealth effect on labour supply. This is what allows a distribution-free aggregate labour block to coexist with a genuinely heterogeneous-agent savings block, but it also means that redistribution has no direct effect on hours — a property that matters for how the working-capital financing income is routed (Section 3.3).*

2.3 Distribution and aggregation

Let $\Phi_t(a, e)$ denote the joint distribution over deposits and efficiency, and $g_{X,t}^a(a, e)$, $g_{X,t}^c(a, e)$ the optimal savings and consumption policies implied by (2.7)–(2.8). The distribution evolves under the standard forward operator

$$\Phi_{t+1}(a', e') = \sum_e \Pi(e, e') \int \mathbf{1}\{g_{X,t}^a(a, e) = a'\} d\Phi_t(a, e), \quad (2.10)$$

implemented numerically by the Young (2010) lottery: an off-grid choice $g^a \in [a_j, a_{j+1}]$ is split between the adjacent nodes with weights $1 - \omega_j$ and $\omega_j = (g^a - a_j)/(a_{j+1} - a_j)$, which preserves the first moment exactly. Aggregates are

$$A_{X,t+1} = \int g_{X,t}^a d\Phi_t, \quad C_{X,t} = \int g_{X,t}^c d\Phi_t. \quad (2.11)$$

2.4 The aggregate closure used in the recursive solution

The heterogeneous-agent block (2.2)–(2.10) is solved exactly at the steady state (it pins the discount factors β_X from deposit-market clearing) and is used for the distributional/welfare overlays. In the recursive global solution, however, the household side is closed at the *aggregate* level by a representative-agent Euler equation — the first rung of a Krusell–Smith ladder, in which the cross-sectional distribution is carried by its mean alone. Three equations define this closure.

(i) Quantity clearing of the deposit market. Aggregate real household deposits equal aggregate bank funding, by construction:

$$A_{X,t} = \frac{\text{dep}_{X,t}}{P_{X,t}^c}. \quad (2.12)$$

(ii) **The aggregate budget constraint.** Consumption is the residual of the household budget, where the gross deposit *claim* carried into t is the bank's gross deposit *obligation* state $P_{X,t}$:

$$C_{X,t} = \underbrace{\frac{P_{X,t}}{P_{X,t}^c}}_{\text{carried wealth}} + \underbrace{\frac{w_{X,t}N_{X,t} + \text{Div}_{X,t} - T_{X,t}^\tau}{P_{X,t}^c}}_{\equiv \text{inc}_{X,t}} + \bar{T}_X - A_{X,t}. \quad (2.13)$$

(iii) **The aggregate deposit Euler equation.** With $x_{X,t} = C_{X,t} - v(N_{X,t})$,

$$x_{X,t}^{-\sigma_X} = \tilde{\beta}_X \mathbb{E}_t \left[(1 + r_{X,t+1}^c) x_{X,t+1}^{-\sigma_X} \right], \quad 1 + r_{X,t+1}^c = (1 + r_{X,t}) \frac{P_{X,t}^c}{P_{X,t+1}^c} \quad (2.14)$$

with $\tilde{\beta}_X = 1/(1 + \bar{r}_X)$, where $\bar{r}_X$ is the steady-state deposit rate. Equation (2.14) is the condition that determines the deposit *rate* $r_{D,t}$ in equilibrium (residual 5 of Section 7.3): quantities clear by (2.12), and the rate is whatever makes households willing to hold exactly what banks fund.

Remark 1 (Two disclosed departures from the primitive environment). The representative-agent stand-in is not the aggregate of (2.2): (a) $\tilde{\beta}_X = 1/(1 + \bar{r}_X)$ exceeds the heterogeneous-agent β_X recovered at the steady state, because precautionary saving in the incomplete-markets block requires $\beta_X(1 + \bar{r}_X) < 1$; and (b) the constant $\bar{T}_X$ in (2.13) is an accounting anchor, calibrated so that the steady state is an exact rest point of the recursive system. It absorbs the working-capital repayment leg, which is netted out of the state $P_{X,t}$ (see equation (4.4)) but accrues to households. Both are stated here rather than buried: the recursive solution is exact in the aggregate but treats the wealth *distribution* as a fixed shape.

3 Production, trade and capital

3.1 Final goods and the terms of trade

The consumption basket of country D aggregates home and foreign goods with an Armington/CES aggregator,

$$C_{D,t} = \left[\varpi^{1/\eta} c_{DD,t}^{\frac{\eta-1}{\eta}} + (1 - \varpi)^{1/\eta} c_{FD,t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad (3.1)$$

with home bias ϖ and trade elasticity η . Normalising the price of the D-good to one and letting p_t be the relative price of the F-good, cost minimisation gives the price index and the import demand schedule

$$P_{D,t}^c = \left[\varpi + (1 - \varpi) p_t^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad IM_{D,t} = (1 - \varpi) \left(\frac{P_{D,t}^c}{p_t} \right)^\eta C_{D,t}, \quad (3.2)$$

$$P_{F,t}^c = \left[\varpi + (1 - \varpi) p_t^{\eta-1} \right]^{\frac{1}{1-\eta}}, \quad IM_{F,t} = (1 - \varpi) (P_{F,t}^c p_t)^\eta C_{F,t}, \quad (3.3)$$

where $P_{F,t}^c$ is expressed in F-good units and the D-good costs $1/p_t$ there. Net exports in own-good units are

$$NX_{D,t} = IM_{F,t} - p_t IM_{D,t}, \quad NX_{F,t} = IM_{D,t} - \frac{IM_{F,t}}{p_t}. \quad (3.4)$$

3.2 Retailers and marginal cost

A unit continuum of retailers buys the homogeneous intermediate good, differentiates it costlessly, and sells under monopolistic competition to a CES final-goods aggregator with elasticity

ϵ_X . Prices are fully flexible, so every retailer charges the constant markup $\epsilon_X/(\epsilon_X - 1)$ over marginal cost and real marginal cost is constant:

$$mc_X = \frac{\epsilon_X - 1}{\epsilon_X}. \quad (3.5)$$

Retail profits $(1 - mc_X)Y_{X,t}$ are rebated lump-sum to households.

3.3 Intermediate producers and the working-capital wedge

The intermediate producer operates a Cobb–Douglas technology in the *predetermined* capital stock and hours,

$$Y_{X,t} = Z_{X,t} K_{X,t}^{\alpha_X} N_{X,t}^{1-\alpha_X}, \quad (3.6)$$

where $K_{X,t}$ was purchased by banks at $t - 1$. Following Bocola’s own open-economy extension (his §V.C), the firm must pre-finance a fraction ζ_X of its wage bill with intra-period bank credit at the gross rate $1 + r_{X,t}^{wc}$. The static problem is

$$\max_{N_{X,t}} mc_X Z_{X,t} K_{X,t}^{\alpha_X} N_{X,t}^{1-\alpha_X} - (1 + \zeta_X r_{X,t}^{wc}) w_{X,t} N_{X,t} - mpk_{X,t} K_{X,t}, \quad (3.7)$$

delivering the factor-demand conditions

$$w_{X,t} = \frac{mc_X(1 - \alpha_X) Y_{X,t}/N_{X,t}}{1 + \zeta_X r_{X,t}^{wc}}, \quad (3.8)$$

$$mpk_{X,t} = mc_X \alpha_X \frac{Y_{X,t}}{K_{X,t}}, \quad (3.9)$$

and a loan demand of

$$L_{X,t} = \zeta_X w_{X,t} N_{X,t}. \quad (3.10)$$

Note that $mpk_{X,t}$ in (3.9) is the marginal product of the *bank-held vintage*: firms rent exactly the stock banks financed one period earlier.

► Equation (3.8) is the **only** channel through which financial spreads reach output on impact. Setting $\zeta_X = 0$ nests the model without it exactly, and doing so collapses the output response to a sovereign-risk shock to approximately zero even when bond prices fall by 30%. The interest cost $\zeta_X r^{wc} w N$ is treated as a real resource cost in the baseline; a switch ($\varsigma^{\text{reb}} \in \{0, 1\}$ in equation (4.21)) can instead rebate it to households, in which case, under GHH, the spread becomes a pure intra-period transfer with no offsetting wealth effect and hours rise with the spread.

3.4 Capital producers, Tobin’s q , and the return on capital

Capital producers convert the final good into installed capital subject to Jermann (1998) adjustment costs. With $\iota_{X,t} \equiv I_{X,t}/K_{X,t}$, the law of motion is

$$K_{X,t+1} = (1 - \delta_X) K_{X,t} + \Phi_X(\iota_{X,t}) K_{X,t}, \quad \Phi_X(\iota) = \gamma_{0,X} \iota^{1-\xi_X} + \gamma_{1,X}, \quad (3.11)$$

with the coefficients pinned so that the adjustment cost is inactive at the steady state ($\Phi_X(\delta_X) = \delta_X$, $\Phi'_X(\delta_X) = 1$):

$$\gamma_{0,X} = \frac{\delta_X^{\xi_X}}{1 - \xi_X}, \quad \gamma_{1,X} = -\frac{\delta_X \xi_X}{1 - \xi_X}. \quad (3.12)$$

The producer solves $\max_{\iota} Q_{X,t} \Phi_X(\iota) K_{X,t} - \iota K_{X,t}$, whose first-order condition delivers marginal Tobin’s q and, inverting (3.11), the investment rate:

$$Q_{X,t} = \frac{1}{\Phi'_X(\iota_{X,t})} = \frac{\iota_{X,t}^{\xi_X}}{\gamma_{0,X}(1 - \xi_X)}, \quad \iota_{X,t} = \left[\frac{K_{X,t+1}/K_{X,t} - (1 - \delta_X) - \gamma_{1,X}}{\gamma_{0,X}} \right]^{\frac{1}{1-\xi_X}}. \quad (3.13)$$

Capital producers' rents, $\Pi_{X,t}^k = Q_{X,t} [K_{X,t+1} - (1 - \delta_X) K_{X,t}] - I_{X,t}$, are rebated to households. The parameter ξ_X is the elasticity of Q with respect to I/K .

The bank holds the capital, so the *realised* gross return between t and $t + 1$ on a unit of capital purchased at t is

$$1 + r_{X,t+1}^k = \frac{mpk_{X,t+1} + (1 - \delta_X) Q_{X,t+1}}{Q_{X,t}} \quad (3.14)$$

which is the object that appears in the intermediary's capital Euler equation.

Total factor productivity in D is carried as a state and follows a deterministic (perfect-foresight) AR(1),

$$\log Z_{D,t+1} = (1 - \rho_z) \log \bar{Z}_D + \rho_z \log Z_{D,t}, \quad (3.15)$$

while $Z_{F,t} = \bar{Z}_F$ throughout: country asymmetries enter through shocks only.

4 Financial intermediaries

This is the core of the model. Each country hosts a representative bank that intermediates *all* of the economy's productive capital, the sovereign bonds held domestically, a cross-border sovereign position, and the working-capital loan book. Its balance sheet is the transmission mechanism from sovereign risk to real activity.

4.1 The balance sheet

At the end of period t the bank in country D holds capital $K_{D,t+1}$ at price $Q_{D,t}$, home sovereign bonds $b_{D,t+1}^D$ at price q_t^D , foreign sovereign bonds $b_{D,t+1}^F$ at price q_t^F converted into D-goods at the terms of trade p_t , and working-capital loans $L_{D,t}$. These are funded by net worth $n_{D,t}$ and deposits $\text{dep}_{D,t}$:

$$\underbrace{Q_{D,t} K_{D,t+1} + q_t^D b_{D,t+1}^D + p_t q_t^F b_{D,t+1}^F + L_{D,t}}_{\equiv \mathcal{A}_{D,t} \text{ (total assets)}} = n_{D,t} + \text{dep}_{D,t}. \quad (4.1)$$

The mirror-image expression for F converts its cross-border leg by $1/p_t$. Bank leverage is $\theta_{X,t} \equiv \mathcal{A}_{X,t}/n_{X,t}$.

4.2 Gross wealth and the deposit-obligation state

Let $h_t \equiv 1 - d_t(1 - \varrho_D)$ denote the survival factor on D-sovereign claims (Section 5). At the start of $t + 1$ the bank collects

$$\mathcal{X}_{D,t+1} = [mpk_{D,t+1} + (1 - \delta_D) Q_{D,t+1}] K_{D,t+1} + \Xi_{t+1}^D b_{D,t+1}^D + p_{t+1} \Xi_{t+1}^F b_{D,t+1}^F, \quad (4.2)$$

where the per-unit bond payoffs are (Section 5)

$$\Xi_{t+1}^D = h_{t+1} [\delta_b + (1 - \delta_b) q_{t+1}^D], \quad \Xi_{t+1}^F = \delta_b + (1 - \delta_b) q_{t+1}^F. \quad (4.3)$$

Against this it owes its predetermined obligation. Because the working-capital loan is extended and repaid *within* the period at the rate locked at t , the obligation the bank carries forward is net of the loan receivable:

$$P_{X,t+1} = (1 + r_{X,t}) \text{dep}_{X,t} - (1 + r_{X,t}^{wc}) L_{X,t} \quad (4.4)$$

so that gross bank wealth at the start of $t + 1$ is simply

$$n_{X,t+1}^g = \mathcal{X}_{X,t+1} - P_{X,t+1}. \quad (4.5)$$

This is exactly the algebraic device that makes P_X a *sufficient* state for the bank's liability side: neither deposits, nor the loan book, nor the loan rate need to be carried separately.

Substituting (4.1) into (4.5) and using the return definitions gives the excess-return representation used throughout:

$$n_{X,t+1}^g = \sum_{j \in \mathcal{J}} (R_{j,t+1} - R_{X,t}) a_{j,t} + R_{X,t} n_{X,t}, \quad R_{X,t} \equiv 1 + r_{X,t}, \quad (4.6)$$

where $\mathcal{J} = \{K, b^D, b^F, L\}$ indexes asset classes, $a_{j,t}$ is the market value of position j , and $R_{j,t+1}$ its gross return. The working-capital leg is riskless as of t : $R_{L,t+1} = 1 + r_{X,t}^{wc}$.

4.3 The banker's problem and the franchise value

Each period a fraction f of bankers exits and pays its net worth out as dividends; the remaining $1 - f$ continue. Entrants receive a startup transfer equal to a fraction ω of the aggregate asset base. A continuing banker maximises the expected discounted value of terminal net worth, discounting with the household stochastic discount factor $\Lambda_{t,t+1}$ of the country it operates in. Writing the value function of a banker with net worth n as $V_t(n)$, the Bellman equation is

$$V_t(n_{X,t}) = \max_{\{a_{j,t}\}} \mathbb{E}_t \left\{ \Lambda_{X,t,t+1} [f n_{X,t+1}^g + (1 - f) V_{t+1}(n_{X,t+1}^g)] \right\}. \quad (4.7)$$

Because (4.6) is linear in n and the constraint below is linear in assets, the value function is linear, $V_t(n) = \varphi_{X,t} n$, where $\varphi_{X,t}$ is the **marginal value of one unit of intermediary net worth** (the franchise value). Defining the bank's **augmented stochastic discount factor**

$$\boxed{\Omega_{X,t,t+1} \equiv \Lambda_{X,t,t+1} [f + (1 - f) \varphi_{X,t+1}]} \quad (4.8)$$

the objective collapses to $\mathbb{E}_t [\Omega_{X,t,t+1} n_{X,t+1}^g]$. The weight on the franchise value is the *survival* probability $1 - f$; the weight f is the exit/payout share.

4.4 The agency friction and the incentive-compatibility constraint

After raising deposits, the banker can abscond with a fraction λ of assets, in which case depositors recover the rest and the bank is liquidated. Deposits are forthcoming only if the franchise is worth at least as much as the divertable proceeds:

$$\boxed{\varphi_{X,t} n_{X,t} \geq \lambda_X \mathcal{A}_{X,t}} \quad (4.9)$$

Following Bocola (2016, eq. 3), *all* asset classes carry the same divertability λ_X — capital, home bonds, cross-border bonds and working-capital loans alike. Equation (4.9) is an endogenous leverage ceiling: $\theta_{X,t} \leq \varphi_{X,t} / \lambda_X$.

Remark 2 (Why a single λ). Allowing $\lambda_K \neq \lambda_{b^D}$ re-opens a portfolio-substitution margin along which a fall in sovereign bond prices *relaxes* the constraint (the bank shifts into the now-cheaper divertability class), which makes sovereign risk expansionary. The single- λ restriction is therefore not a simplification but a structural choice inherited from Bocola.

4.5 First-order conditions and the closed-form multiplier

Attach the multiplier $\tilde{\mu}_{X,t} \geq 0$ to (4.9) and write the Lagrangian of the banker's problem as

$$\mathcal{L} = (1 + \tilde{\mu}_{X,t}) \mathbb{E}_t [\Omega_{X,t,t+1} n_{X,t+1}^g] - \tilde{\mu}_{X,t} \lambda_X \mathcal{A}_{X,t}, \quad (4.10)$$

using $V_t(n) = \mathbb{E}_t[\Omega n_{t+1}^g]$ on the left of (4.9). It is convenient to rescale the multiplier onto the unit interval,

$$\mu_{X,t} \equiv \frac{\tilde{\mu}_{X,t}}{1 + \tilde{\mu}_{X,t}} \in [0, 1]. \quad (4.11)$$

Differentiating (4.10) with respect to each position $a_{j,t}$ and using (4.6) gives the **unified asset-pricing condition**

$$\boxed{\mathbb{E}_t \left[\Omega_{X,t,t+1} (R_{j,t+1} - R_{X,t}) \right] = \lambda_X \mu_{X,t}, \quad j \in \{K, b^D, b^F, L\}} \quad (4.12)$$

together with the Karush–Kuhn–Tucker complementarity

$$\mu_{X,t} \geq 0, \quad \text{slack}_{X,t} \equiv \varphi_{X,t} n_{X,t} - \lambda_X \mathcal{A}_{X,t} \geq 0, \quad \mu_{X,t} \cdot \text{slack}_{X,t} = 0. \quad (4.13)$$

Every asset class earns the *same* shadow excess return $\lambda_X \mu_{X,t}$: the constraint prices scarce intermediary equity, not asset-specific risk.

Proposition 1 (Franchise value and the closed-form multiplier; Bocola 2016, Prop. 1). *At an optimum the franchise value satisfies the recursion*

$$\varphi_{X,t} = \frac{\mathbb{E}_t[\Omega_{X,t,t+1}] R_{X,t}}{1 - \mu_{X,t}}, \quad (4.14)$$

and, whenever the incentive constraint binds, the multiplier admits the closed form

$$\mu_{X,t} = \max \left\{ 1 - \frac{\mathbb{E}_t[\Omega_{X,t,t+1}] R_{X,t} n_{X,t}}{\lambda_X \mathcal{A}_{X,t}}, 0 \right\}. \quad (4.15)$$

Proof. Evaluate the objective at the optimum using (4.6) and (4.12):

$$\begin{aligned} \varphi_{X,t} n_{X,t} &= \mathbb{E}_t[\Omega n_{t+1}^g] = \sum_j \mathbb{E}_t[\Omega (R_{j,t+1} - R_{X,t})] a_{j,t} + \mathbb{E}_t[\Omega] R_{X,t} n_{X,t} \\ &= \mu_{X,t} \lambda_X \mathcal{A}_{X,t} + \mathbb{E}_t[\Omega] R_{X,t} n_{X,t}. \end{aligned}$$

If the constraint binds, $\lambda_X \mathcal{A}_{X,t} = \varphi_{X,t} n_{X,t}$, so $\varphi_{X,t} n_{X,t} (1 - \mu_{X,t}) = \mathbb{E}_t[\Omega] R_{X,t} n_{X,t}$, which is (4.14); substituting (4.14) back into the binding constraint and solving for μ gives (4.15). If the constraint is slack, $\mu_{X,t} = 0$ and (4.14) still holds, because the excess-return terms vanish by (4.12). The $\max\{\cdot, 0\}$ operator is exactly the complementarity (4.13). $\square$

► Equation (4.15) is the analytical heart of the transmission mechanism. A fall in sovereign bond prices reduces $n_{X,t}$ through (4.5), which mechanically raises $\mu_{X,t}$: scarcer equity, a tighter constraint, higher required excess returns on every asset class, and — via (4.18) — a higher cost of working capital and lower hours. The premium is endogenous and requires no ad hoc spread rule.

4.6 The three pricing conditions in explicit form

Specialising (4.12) to each class yields the equations imposed in equilibrium.

Capital. With $R_{K,t+1} = 1 + r_{X,t+1}^k$ from (3.14):

$$\mathbb{E}_t \left[\Omega_{X,t,t+1} (r_{X,t+1}^k - r_{X,t}) \right] = \lambda_X \mu_{X,t}. \quad (4.16)$$

This is the equation that pins down the capital stock $K_{X,t+1}$ carried into the next period (residuals 1–2 of Section 7.3).

Sovereign bonds. With $R_{b,t+1} = \Xi_{t+1}/q_t$, condition (4.12) rearranges into an explicit pricing formula,

$$q_t^D = \frac{\mathbb{E}_t[\Omega_{D,t,t+1} \Xi_{t+1}^D]}{\mathbb{E}_t[\Omega_{D,t,t+1}] R_{D,t} + \lambda_D \mu_{D,t}}, \quad q_t^F = \frac{\mathbb{E}_t[\Omega_{F,t,t+1} \Xi_{t+1}^F]}{\mathbb{E}_t[\Omega_{F,t,t+1}] R_{F,t} + \lambda_F \mu_{F,t}}. \quad (4.17)$$

The denominator makes the *liquidity* (constraint) component of the yield explicit: even a default-free bond trades below its risk-neutral present value by the factor $\lambda\mu$. The numerator carries the default risk.

Working capital. The loan rate is set at t and is riskless, so $\mathbb{E}_t[\Omega](1 + r_{X,t}^{wc} - R_{X,t}) = \lambda_X \mu_{X,t}$, i.e.

$$\boxed{r_{X,t}^{wc} = r_{X,t} + \underbrace{\frac{\lambda_X \mu_{X,t}}{\mathbb{E}_t[\Omega_{X,t,t+1}]}_{\text{lending spread}}} \quad (4.18)$$

Equation (4.18) is the model's **credit spread**. It is the single quantity that links the financial block to the real block, through (3.8).

4.7 Net worth accumulation, entry and exit

Aggregating over bankers, net worth carried out of period t is the retained wealth of surviving bankers plus the endowment of entrants:

$$\boxed{n_{X,t} = (1 - f) n_{X,t}^g + \omega_X \mathcal{A}_{X,t}} \quad (4.19)$$

with $n_{X,t}^g = \mathcal{X}_{X,t} - P_{X,t}$ from (4.5), and the corresponding net payout to households is

$$\text{div}_{X,t}^{\text{bank}} = f n_{X,t}^g - \omega_X \mathcal{A}_{X,t}. \quad (4.20)$$

Deposits are then the residual funding need, $\text{dep}_{X,t} = \mathcal{A}_{X,t} - n_{X,t}$, and the obligation state evolves by (4.4). Total dividends received by households consolidate retail profits, capital-producer rents, and the banking sector:

$$\text{Div}_{X,t} = (1 - mc_X) Y_{X,t} + \Pi_{X,t}^k + \text{div}_{X,t}^{\text{bank}} + \varsigma^{\text{reb}} \zeta_X r_{X,t}^{wc} \omega_{X,t} N_{X,t}, \quad (4.21)$$

where $\varsigma^{\text{reb}} \in \{0, 1\}$ is the working-capital rebate switch discussed in Section 3.3 (baseline: $\varsigma^{\text{reb}} = 0$).

4.8 Steady-state calibration of the agency friction

At the steady state the constraint binds with equality and the system (4.14)–(4.15) collapses to a scalar fixed point. Given a target leverage $\bar{\theta}$ and a target credit spread $\bar{\sigma} \equiv \bar{r}^k - \bar{r}$, the pair (λ_X, ω_X) is solved from

$$\bar{\varphi} = \frac{\bar{\Omega}(1 + \bar{r})}{1 - \bar{\mu}}, \quad \bar{\Omega} = \beta^{\text{int}} [f + (1 - f)\bar{\varphi}], \quad \bar{\mu} = \frac{\bar{\Omega} \bar{\sigma} \bar{\theta}}{\bar{\varphi}}, \quad (4.22)$$

$$\lambda_X = \frac{\bar{\varphi}}{\bar{\theta}}, \quad \omega_X = \frac{\mathcal{D}}{\bar{\theta}} - (1 - f) \bar{\sigma}, \quad \mathcal{D} \equiv 1 - (1 - f)(1 + \bar{r}), \quad (4.23)$$

where β^{int} is the bankers' discount factor, set to $1/(1 + \bar{r})$ so that $\beta^{\text{int}} R = 1$ at the steady state. The scalar fixed point (4.22) can *fold*: for high spread/leverage combinations the map has no low root and the calibration is infeasible. The solver selects the least root, which is also the one value iteration from below converges to.

Two internal consistency conditions define the steady state of the bank block: the level of net worth implied by the binding constraint, $\bar{n}^{IC} = \lambda \bar{A} / \bar{\varphi}$, must equal the level implied by the accumulation identity (4.19),

$$\bar{n}^{ACC} = \frac{1}{\mathcal{D}} \sum_{j \in \mathcal{J}} \left[(1 - f)(\bar{R}_j - \bar{R}) + \omega_X \right] \bar{a}_j, \quad (4.24)$$

and this equality is what pins the steady-state return on capital $\bar{r}_X^k$ in the first stage of the steady-state solve.

5 Sovereign risk and the default event

5.1 Debt instrument

Both governments issue Hatchondo–Martinez/Chatterjee–Eyigungor perpetuities: a unit of the stock pays a coupon δ_b this period and $(1 - \delta_b)$ units of the same bond survive into the next. Duration is approximately $1/\delta_b$ quarters. The ex-coupon payoff to a unit of stock is Ξ_{t+1} of equation (4.3).

► *Long duration is essential, not cosmetic. Because the price q_t capitalises the entire expected future path of default risk, a persistent increase in π^d produces a large mark-to-market loss on bank balance sheets. Shortening duration to $\delta_b = 0.25$ cuts the repricing by roughly a factor of six and reverses the sign of the output response.*

5.2 The exogenous, priced risk process

Following Bocola (2016, eqs. 11–12), default risk is **exogenous**: it is a priced probability, never a function of the debt stock or of any endogenous state. A latent risk factor s_t follows

$$s_{t+1} = (1 - \rho_s) \bar{s} + \rho_s s_t + \sigma_s \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, 1), \quad (5.1)$$

and the one-quarter-ahead priced probability of default is the logistic transform

$$\pi_t^d = \pi^d(s_t) = \frac{1}{1 + e^{-s_t}}, \quad \bar{s} = \log \frac{0.001}{0.999} \text{ so that } \pi^d(\bar{s}) = 0.1\%. \quad (5.2)$$

Two distinct objects must be kept separate:

- the **priced** probability π_t^d , which enters bond pricing (4.17) and every expected-return condition (4.12);
- the **realised** indicator $d_t \in \{0, 1\}$, which enters realised payoffs (4.2) and the government's flow budget.

The baseline experiment sets $\pi_t^d > 0$ with $d_t \equiv 0$: risk is priced but never realised. This is precisely Bocola's "pass-through" design — the entire contraction is generated by the *anticipation* of an event that does not occur.

5.3 The default event

The feared event is a single deterministic pure haircut: a write-down of the whole D-claim to a recovery value ϱ_D , applied to coupon and continuation value alike, so the survival factor is

$$h_t = 1 - d_t (1 - \varrho_D), \quad \varrho_D = 0.45 \text{ (the 2012 Greek PSI; Bocola's } D = 0.55). \quad (5.3)$$

There are no exogenous scarring costs: no output cost of default, no capital-quality shock, no bank recapitalisation. The recession in the default state arises *endogenously*, through the same balance-sheet mechanism (4.5)–(4.15) operating on a smaller asset base. The F-sovereign never defaults, so F-bonds are safe in both states.

5.4 The expectation operator

The conditional expectations in (4.8), (4.12), (4.17) and (2.14) are genuine multi-dimensional integrals over (i) the innovation to s and (ii) the discrete default fork. Let $\{\varepsilon_m, w_m\}_{m=1}^M$ be the Gauss–Hermite nodes and weights of the probabilists’ Hermite rule, so that $s_{t+1}^{(m)} = (1 - \rho_s)\bar{s} + \rho_s s_t + \sigma_s \varepsilon_m$, and let $d' \in \{0, 1\}$ index the regime. For any function g of next period’s state,

$$\mathbb{E}_t[g] = \sum_{m=1}^M w_m \left\{ (1 - \pi_t^{d'}) g(0, \mathcal{S}_{t+1}^{(m)}) + \pi_t^{d'} \left[\varkappa_{\text{tpi}} g^{\text{hon}}(\mathcal{S}_{t+1}^{(m)}) + (1 - \varkappa_{\text{tpi}}) g(1, \mathcal{S}_{t+1}^{(m)}) \right] \right\} \quad (5.4)$$

where $g(d', \cdot)$ evaluates the equilibrium decision rules of regime d' at a *reachable* next-period state — never a frozen stand-in economy — and g^{hon} is the backstop-honoured branch of Section 6.4. Setting $\varkappa_{\text{tpi}} = 0$ collapses (5.4) to the two-branch (no-backstop) case, and $\pi^{d'} \equiv 0$ nests the risk-neutral model exactly.

5.5 Branch stochastic discount factors and the endogenous premium

The banker discounts with the household kernel of its own country, evaluated branch-by-branch on the GHH composite:

$$\Lambda_{X,t,t+1}^{(d')} = \beta_X^{\text{int}} \left(\frac{x_{X,t}}{x_{X,t+1}^{(d')}} \right)^{\sigma_X}, \quad \Omega_{X,t,t+1}^{(d')} = \Lambda_{X,t,t+1}^{(d')} \left[f + (1 - f) \varphi_{X,t+1}^{(d')} \right]. \quad (5.5)$$

Because consumption is lower in the default branch, $x^{(1)} < x^{(0)}$ and hence $\Omega^{(1)} > \Omega^{(0)}$: the bank values wealth more in the bad state. This is what makes the bond carry a genuine *risk premium* rather than a pure actuarial discount, and it is what makes $\mathbb{E}_t[\Omega]$ in (4.15) *fall* when risk rises — tightening the constraint.

► A constant discount factor on the no-default branch (i.e. $\Lambda^{(0)} \equiv \beta^{\text{int}}$) cannot fall when risk rises, so $\mathbb{E}_t[\Omega]$ climbs with φ' and the constraint counterfactually loosens. The state-contingent kernel (5.5) is required for the constraint to tighten with risk.

The bond price (4.17) then decomposes into three pieces:

$$q_t^D = \underbrace{\frac{\mathbb{E}_t[\Xi_{t+1}^D]}{R_{D,t}}}_{\text{expected payoff}} + \underbrace{\frac{\text{Cov}_t(\Omega, \Xi_{t+1}^D)}{\mathbb{E}_t[\Omega] R_{D,t}}}_{\text{risk premium } (<0)} - \underbrace{\frac{\lambda_D \mu_{D,t}}{\mathbb{E}_t[\Omega] R_{D,t} + \lambda_D \mu_{D,t}}}_{\text{liquidity/constraint discount}} q_t^D. \quad (5.6)$$

6 Government and policy

6.1 The consolidated government budget constraint

The government of country X finances an exogenous expenditure stream G_X , coupon payments on the surviving stock, and any bailout outlay, with lump-sum taxes and new issuance. In own-good units,

$$G_X + \delta_b B_{X,t} h_t = T_{X,t}^\tau + q_t^X \left[B_{X,t+1} - (1 - \delta_b) B_{X,t} h_t \right] \quad (6.1)$$

so that the debt stock evolves as

$$B_{X,t+1} = (1 - \delta_b) B_{X,t} h_t + \frac{G_X + \delta_b B_{X,t} h_t - T_{X,t}^\tau}{q_t^X}. \quad (6.2)$$

Because q_t^X falls when risk rises, (6.2) contains the standard *financing* amplification: the same primary deficit requires more face value to be placed when the sovereign is under stress. The end-of-period stock is forward-integrated *inside* every equilibrium evaluation and absorbed by banks; clearing against a fixed $\bar{B}$ instead opens a Walras leak of order 0.5% of GDP per 5% debt deviation.

6.2 The fiscal rule

Taxes follow a Bohn (1998) rule in the form Bocola estimates, i.e. as a constant *elasticity* of the tax level with respect to the surviving debt stock:

$$T_{X,t}^\tau = \bar{T}_X^\tau \left(\frac{B_{X,t} h_t}{\bar{B}_X} \right)^{\gamma_\tau}, \quad \gamma_\tau = 1 \quad (6.3)$$

with the steady-state anchor $\bar{T}_X^\tau = G_X + \delta_b \bar{B}_X (1 - \bar{q}^X)$ obtained from (6.1) at a constant stock. *Remark 3* (Elasticity, not level). Reading γ_τ as a coefficient on the debt *level* rather than as an elasticity turns a 55% haircut into a fiscal windfall on the order of 50% of GDP, which makes the default event expansionary and renders most of the $d = 1$ region unsolvable. Under (6.3) a haircut to $\varrho_D = 0.45$ delivers $B'/\bar{B} = 0.451$, as it should: the fiscal-relief leg of default is real but second-order relative to the bank-loss leg at the calibrated exposure.

6.3 The monetary block of the union

Prices are fully flexible and there is no nominal rigidity, so the model contains no Taylor rule and no nominal interest rate to set. What a monetary union *does* impose on real allocations is modelled directly, in two parts.

(i) A single union-wide funding market. The two national deposit-market clearing conditions are replaced by one union-wide clearing condition (in D-good units) together with a no-arbitrage condition across national deposit legs. Deposits are own-good claims remunerated at national rates, and a frictionless union interbank market equalises their real returns:

$$(1 + r_{D,t}) = (1 + r_{F,t}) \frac{\mathbb{E}_t[p_{t+1}]}{p_t} - \kappa_{\text{nfa}} \frac{P_{D,t} - P_{F,t}}{\bar{Y}_D} \quad (6.4)$$

This **deposit-UIP** condition is the flexible-price image of a single nominal policy rate plus national inflation differentials, i.e. the Backus–Kehoe–Kydland / Baxter–Crucini single-traded-bond margin. Together with (2.14) for D it determines both national deposit rates. Because pass-through through the union interbank is zero-profit, the cross-border deposit position is the absorption margin that breaks the national saving-equals-investment trap; imposing the literal condition $r_{D,t} = r_{F,t}$ on own-good legs instead is *incorrect*, because it leaves an unassigned real-exchange-rate valuation profit and opens a Walras leak.

(ii) A stationarity-inducing external premium. The last term of (6.4) is the Schmitt-Grohé–Uribe debt-elastic premium on the net external position (Bocola’s only foreign friction), proxied here by the cross-country wealth imbalance $P_{D,t} - P_{F,t}$. It is exactly zero at the symmetric steady state, so it is undistorting there and only induces stationarity off-steady-state. $\kappa_{\text{nfa}} = 0$ nests frictionless parity.

6.4 The TPI/OMT backstop

The union’s central bank operates a sovereign backstop of the OMT/TPI type. Two formulations are implemented; both are *announcement*, not rate, instruments.

(a) Priced activation (the production formulation). The default fork of (5.4) splits into a **backstop-honoured** and a **backstop-reneged** branch, weighted by the priced activation probability $\varkappa_{\text{tpi}} \in [0, 1]$. In the honoured branch the central bank (i) redeems the D-bond at a near-par value ϱ^{tpi} and (ii) averts a fraction ς of the recession, so that the honoured continuation is a blend of the two regimes' decision rules:

$$g^{\text{hon}}(\mathcal{S}') = (1 - \varsigma) g(1, \mathcal{S}') + \varsigma g(0, \mathcal{S}'), \quad (6.5)$$

$$\Xi_{t+1}^{\text{D,hon}} = \varrho^{\text{tpi}} \left[\delta_b + (1 - \delta_b) q_{t+1}^{\text{D,hon}} \right], \quad \Xi_{t+1}^{\text{D,ren}} = \varrho_{\text{D}} \left[\delta_b + (1 - \delta_b) q_{t+1}^{\text{D,(1)}} \right]. \quad (6.6)$$

At $\varrho^{\text{tpi}} = \varsigma = 1$ a honoured backstop fully neutralises the event, so the effective default probability becomes $\pi_t^d(1 - \varkappa_{\text{tpi}})$ — the textbook reading that a fully credible backstop removes the risk premium at zero purchases. At $\varkappa_{\text{tpi}} = 0$ or $\varsigma = 0$ the two-branch solve is nested exactly.

(b) A mechanical price floor (the Markov-switching formulation). In the regime $\mathfrak{s}_t^{\text{tpi}} = 1$ the central bank supports the D-bond at a floor that prices the *same* expected payoff at the steady-state constraint spread, purchasing quantity $\mathcal{Q}_t^{\text{cb}}$ according to a portfolio-balance rule:

$$q_t^{\text{floor}} = \frac{\mathbb{E}_t[\Xi_{t+1}^{\text{D}}]}{1 + r_{\text{D},t} + \bar{\lambda}\bar{\mu}/\bar{\Omega}}, \quad \mathcal{Q}_t^{\text{cb}} = \frac{\max\{0, q_t^{\text{floor}} - q_t^{\text{D,free}}\}}{\psi_{\text{cb}}} \mathfrak{s}_t^{\text{tpi}}, \quad q_t^{\text{D}} = q_t^{\text{D,free}} + \psi_{\text{cb}} \mathcal{Q}_t^{\text{cb}}. \quad (6.7)$$

The floor strips out the liquidity component of the yield while leaving default compensation intact. Any central-bank profit or loss is rebated to the fiscal authority, so (6.1) continues to hold as a consolidated constraint. The max operator in (6.7) is smoothed in implementation, since it is non-differentiable exactly where the floor switches on.

6.5 Macroprudential policy

The model contains **no macroprudential rule**: this is a deliberate scope decision, not an omission of an implemented block. The natural policy object, were one to be added, is the divertability parameter λ_X of (4.9), equivalently the leverage ceiling

$$\theta_{X,t} \leq \bar{\theta}_{X,t} = \frac{\varphi_{X,t}}{\lambda_X}, \quad (6.8)$$

which a capital requirement would tighten. The comparative statics of such a policy are fully characterised in closed form by (4.22): a permanently tighter λ lowers steady-state leverage one-for-one and, through $\bar{\mu} = \bar{\Omega}\bar{\sigma}\bar{\theta}/\bar{\varphi}$, raises the steady-state credit spread (4.18) — trading a smaller crisis amplification for a permanently higher cost of capital. Making λ state-contingent (e.g. a countercyclical buffer $\lambda_t = \bar{\lambda} \exp\{-\phi_{\text{mp}} \log(n_{X,t}/\bar{n}_X)\}$) would enter the system only through (4.15) and would require no other change.

7 Equilibrium

7.1 Market clearing

Goods markets. Each country's good clears against domestic absorption and net exports,

$$Y_{\text{D},t} = P_{\text{D},t}^c C_{\text{D},t} + I_{\text{D},t} + G_{\text{D}} + NX_{\text{D},t}, \quad (7.1)$$

$$Y_{\text{F},t} = P_{\text{F},t}^c C_{\text{F},t} + I_{\text{F},t} + G_{\text{F}} + NX_{\text{F},t}. \quad (7.2)$$

The term $P_{X,t}^c C_{X,t}$ is consumption expenditure valued in own-good units; subtracting imports and adding exports through NX recovers the own-good content. By Walras' law, (7.2) and the current-account identity are implied by the remaining conditions and are therefore *dropped* from the solved system and monitored as diagnostics.

Deposit market (union-wide). Household saving equals bank funding, by quantity, with the union interbank reallocating across countries:

$$A_{D,t} P_{D,t}^c + p_t A_{F,t} P_{F,t}^c = \text{dep}_{D,t} + p_t \text{dep}_{F,t}, \quad A_{X,t} = \frac{\text{dep}_{X,t}}{P_{X,t}^c}, \quad (7.3)$$

with the cross-border deposit position $\text{nfa}_{D,t}^{\text{dep}}$ as the absorption margin, and the rate structure pinned by (2.14) and (6.4).

Sovereign bond markets. Banks absorb the entire outstanding stock of each sovereign:

$$b_{D,t+1}^D + b_{F,t+1}^D = B_{D,t+1}, \quad b_{F,t+1}^F + b_{D,t+1}^F = B_{F,t+1}. \quad (7.4)$$

The cross-border split is held at its calibrated share in the projection solver (b_F^D/B_D fixed at the EBA-based exposure); in the path-based implementation it is instead determined by the cross-border FOC with quadratic portfolio adjustment costs,

$$b_{D,t}^F = \bar{b}_D^F + \frac{1}{\psi^b} \left[\mathbb{E}_t [R_{t+1}^F] - R_{D,t} - \overline{\text{exc}} - \lambda_D \mu_{D,t} / \mathbb{E}_t [\Omega_D] \right]. \quad (7.5)$$

Capital and labour. Capital market clearing is implicit in the intermediary's balance sheet — banks hold the entire capital stock, and the capital Euler (4.16) determines $K_{X,t+1}$. Labour clears at the wage satisfying (2.9) and (3.8) jointly.

External account. The current account identity

$$NX_{D,t} + (\text{net cross-border factor income}) = \Delta(\text{net foreign asset position}) \quad (7.6)$$

holds as an implication and is monitored as a Walras diagnostic.

7.2 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium). Let the aggregate state be

$$\mathcal{S}_t = (K_{D,t}, K_{F,t}, P_{D,t}, P_{F,t}, B_{D,t}, s_t, Z_{D,t}) \in \mathbb{R}^7, \quad d_t \in \{0, 1\}, \quad (7.7)$$

i.e. the two capital stocks, the two banks' gross deposit obligations, the risky debt stock, the sovereign-risk factor and TFP, together with the discrete default regime. A recursive competitive equilibrium consists of

1. decision rules $\{N_X(d, \mathcal{S}), K'_X(d, \mathcal{S}), r_X(d, \mathcal{S}), p(d, \mathcal{S})\}$;
2. valuation rules $\{\varphi_X(d, \mathcal{S}), q^X(d, \mathcal{S})\}$ and household aggregates $\{C_X(d, \mathcal{S}), A_X(d, \mathcal{S})\}$;
3. multipliers $\mu_X(d, \mathcal{S}) \geq 0$ and a transition law $\mathcal{S}' = \mathcal{T}(d, \mathcal{S}, d', \varepsilon')$;

such that, at every $(d, \mathcal{S})$: households optimise ((2.9), (2.14)) subject to (2.13); firms optimise ((3.8)–(3.9)) and capital producers satisfy (3.13); intermediaries optimise ((4.12), (4.14)) subject to the incentive constraint (4.9) with complementarity (4.13); the government satisfies (6.1)–(6.3); all markets clear ((7.1), (7.3), (7.4), (6.4)); and expectations are formed under (5.4) with $\mathcal{S}'$ generated by the same rules.

Table 2: The seven market-clearing conditions and the unknown each pins down.

#	Condition	Equation	Pins
1	Capital Euler, D	$\mathbb{E}[\Omega_D(r_D^{k'} - r_D)] - \lambda_D \mu_D = 0$	K'_D
2	Capital Euler, F	$\mathbb{E}[\Omega_F(r_F^{k'} - r_F)] - \lambda_F \mu_F = 0$	K'_F
3	Labour market, D	$\chi_D N_D^{1/\nu_D} - w_D / P_D^c = 0$	N_D
4	Labour market, F	$\chi_F N_F^{1/\nu_F} - w_F / P_F^c = 0$	N_F
5	Household deposit Euler, D	$x_D^{-\sigma} - \tilde{\beta}_D \mathbb{E}[(1 + r_D^c) x_D'^{-\sigma}] = 0$	r_D
6	Deposit-UIP (union)	$(1 + r_D) - (1 + r_F) \mathbb{E}[p'] / p + \kappa_{\text{nfa}} \Delta P / \bar{Y} = 0$	r_F
7	Goods market, D	$Y_D - P_D^c C_D - I_D - N X_D - G_D = 0$	p

Dropped (Walras-redundant), monitored as diagnostics: goods market F; current account.

7.3 The solved system, equation by equation

At each point $(d, \mathcal{S})$ the equilibrium is characterised by seven equations in seven unknowns,

$$\mathbf{x} = (N_D, N_F, K'_D, K'_F, r_D, r_F, p), \quad (7.8)$$

listed in Table 2. The valuations φ_X , q^X and the multiplier μ_X are *read off* the recursions (4.14), (4.17) and (4.15) given the continuation, rather than solved for — which is what breaks the simultaneity between the bond price and next period's state.

7.4 Definitions of the key endogenous variables

The objects reported in the experiments are defined as follows.

$$\text{Credit / lending spread (annualised bp)} \quad \text{sp}_{X,t} = 4 \times 10^4 \cdot \frac{\lambda_X \mu_{X,t}}{\varphi_{X,t}}, \quad (7.9)$$

$$\text{Bank leverage} \quad \theta_{X,t} = \frac{\mathcal{A}_{X,t}}{n_{X,t}} \leq \frac{\varphi_{X,t}}{\lambda_X}, \quad (7.10)$$

$$\text{Sovereign yield / spread over the safe rate} \quad y_t^D = \frac{\delta_b + (1 - \delta_b) q_t^D}{q_t^D} - 1, \quad \text{spr}_t^D = y_t^D - y_t^F, \quad (7.11)$$

$$\text{Constraint slack} \quad \text{slack}_{X,t} = \varphi_{X,t} n_{X,t} - \lambda_X \mathcal{A}_{X,t} \geq 0, \quad (7.12)$$

$$\text{Real exchange rate} \quad \text{rer}_t = \frac{p_t P_{F,t}^c}{P_{D,t}^c}, \quad \text{Debt/GDP} = \frac{q_t^D B_{D,t}}{4Y_{D,t}}. \quad (7.13)$$

7.5 Decomposition of the output response

Because production (3.6) is Cobb–Douglas and capital is predetermined, the impact response of output is an exact identity in three terms:

$$d \log Y_{D,t} = d \log Z_{D,t} + \alpha_D d \log K_{D,t} + (1 - \alpha_D) d \log N_{D,t}. \quad (7.14)$$

Substituting the labour-market condition (2.9) together with (3.8) and solving for hours,

$$d \log N_{D,t} = \Theta \left[d \log Z_{D,t} + \alpha_D d \log K_{D,t} - \underbrace{d \log (1 + \zeta_D r_{D,t}^{wc})}_{\text{financial wedge}} - d \log P_{D,t}^c \right], \quad \Theta = \frac{1}{1/\nu_D + \alpha_D} \quad (7.15)$$

and, using (4.18), the financial wedge splits additively into a **deposit-rate** leg and a **credit-spread** leg, $r^{wc} = r_D + \lambda_D \mu_D / \mathbb{E}[\Omega_D]$. The second is the Bocola channel: it is the only term through which an increase in the priced probability of sovereign default reaches output.

8 The steady state

The deterministic steady state is imposed to be **symmetric**: country asymmetries enter through shocks only, so $p_{ss} = 1$ and every real quantity is common across countries. It is computed in two stages.

Stage 1: $\{\bar{r}_D^k, \bar{r}_F^k, \bar{p}\}$. Two capital-market conditions — the equality of $\bar{n}^{IC}$ and $\bar{n}^{ACC}$ from Section 4.8, one per country — plus external balance,

$$NX_D + (\text{net cross-border coupon income}) = 0, \quad (8.1)$$

determine the two returns on capital and the terms of trade. TFP is then rescaled to normalise $\bar{Y}_X = 1$ and $\bar{N}_X = 1$, and χ_X is set from the GHH labour condition (2.9).

Stage 2: $\{\beta_D, \beta_F\}$. The heterogeneous-agent block is solved for each candidate discount factor and the deposit-market condition

$$A_X^{ss}(\beta_X) = (\bar{\theta}_X - 1)\bar{n}_X \quad (8.2)$$

is inverted for β_X . Under the symmetric-steady-state doctrine $\beta_F = \beta_D$ and the F condition is verified rather than imposed — at a symmetric steady state the union clearing coincides with each national market and the cross-border deposit position is zero.

At the calibrated steady state the incentive constraint **binds** ($\bar{\mu} \approx 0.02$, slack = 0), which is a deliberate choice: Bocola’s own posterior puts $\bar{\mu} \approx 0.001$, i.e. exactly on the KKT kink, where any shock drives some period’s multiplier across zero and the solver loses a valid Jacobian. The Gertler–Karadi (2011) robustly-binding calibration (leverage 4, a 200 bp annual spread) keeps the constraint strictly binding through both experiments at the cost of smaller — but correctly signed — amplification.

9 Numerical solution

The model is solved **globally**, as recursive decision rules over the seven continuous states (7.7) and the discrete regime d , with separate coefficient sets per regime.

Approximation. Every rule is a Chebyshev polynomial on a Smolyak sparse grid built from nested Chebyshev-extrema point sets (Krueger–Kubler): the grid is the union of tensor products of per-level “new point” sets over multi-indices i with $\sum_j (i_j - 1) \leq \ell$, and the basis uses the same index set over “new degree” sets, so the collocation matrix is square and interpolation is exact at the nodes. Anisotropic levels ℓ_j concentrate resolution on the risk factor s and the two net-worth states, where the constraint boundary moves. Rules that must remain positive are collocated in logs; interest rates are carried as gross rates $1 + r$.

Algorithm (time iteration). Each sweep freezes the previous iterate as the continuation, solves the seven residuals of Table 2 pointwise by Newton’s method at every grid node and in both regimes, reads the valuations off (4.14)–(4.15), then damps and refits the coefficients. Convergence requires *both* a settled rule and every point clearing; nodes that fail to clear retain their previous values and are down-weighted in the fit, so a failed corner cannot poison the global basis.

Smoothing of kinks. Guards that would otherwise plant a plateau inside a fitted object — the caps on μ and φ , the net-worth floor of (4.19), the max in (6.7) — are replaced by the smooth counterparts $\text{smax}(x, \underline{x}) = \underline{x} + \frac{1}{2}[(x - \underline{x}) + \sqrt{(x - \underline{x})^2 + \epsilon^2}]$. The KKT switch in (4.15) is *not* smoothed: it is the complementarity itself, not a numerical guard.

Known limitations. The magnitudes reported at the baseline are indicative rather than estimated: the solve converges at approximation level $\ell = 1$; $\ell = 2$ does not converge, because the near-unit-root corners of the $d' = 1$ regime poison the global basis. The heterogeneous-agent distribution is carried by its mean (Remark 1), the household deposit Euler does not price the default branch (households' deposits are riskless, faithful to Bocola), and the cross-border bond split is fixed at its calibrated share in the projection solver.

10 Calibration

A Symbol-to-code map

B Summary of the transmission mechanism

For reference, the chain of equations through which an exogenous increase in the priced probability of sovereign default contracts real activity:

1. $s_t \uparrow$ raises π_t^d through (5.2);
2. the multi-branch expectation (5.4) places more weight on the haircut payoff $\Xi^{\text{D,ren}}$, so q_t^{D} falls through (4.17) — amplified by duration $1/\delta_b$;
3. bank gross wealth $n_{\text{D}}^g = \mathcal{X}_{\text{D}} - P_{\text{D}}$ falls in (4.5) (mark-to-market loss), so n_{D} falls in (4.19);
4. simultaneously $\mathbb{E}_t[\Omega_{\text{D}}]$ falls, because the state-contingent kernel (5.5) puts weight on a low-consumption branch;
5. both effects raise $\mu_{\text{D},t}$ in the closed form (4.15): the incentive constraint tightens;
6. the credit spread $\lambda_{\text{D}}\mu_{\text{D}}/\mathbb{E}[\Omega_{\text{D}}]$ widens (4.18), raising the cost of working capital;
7. the effective wage rises in (3.8), hours fall through (2.9), and output falls by (7.15);
8. $\mu_{\text{D}} \uparrow$ also raises the required return on capital (4.16), so K'_{D} and investment fall, propagating the contraction into subsequent periods;
9. the government must place more face value at a lower price (6.2), taxes rise through (6.3), and the loop repeats.

No default is ever realised: $d_t \equiv 0$ throughout. The entire contraction is the pass-through of *priced* risk.

Table 3: Baseline quarterly calibration. Symmetric across countries unless noted.

Parameter	Symbol	Value	Source / target
<i>Households</i>			
Inverse EIS	σ	1.0	Bocola §II.A.1 (log utility)
Frisch elasticity	ν	2.0	Bocola Table 1 ($1/\nu = 0.5$)
Persistence of idiosyncratic risk	ρ_e	0.90	standard
S.d. of idiosyncratic risk	σ_e	0.20	standard
Borrowing limit	$\underline{a}$	0	no borrowing
<i>Production and capital</i>			
Capital share	α	0.30	Bocola Table 1
Depreciation	δ	0.025	standard
Elasticity of q to I/K	ξ	0.42	Bocola Table 2 (posterior mean)
Demand elasticity	ϵ	6.0	20% markup
Working-capital share	ζ	1.0	Neumeyer–Perri; Bocola §V.C
<i>Intermediaries</i>			
Exit/payout share	f	0.04	Bocola Table 2 ($\psi = 0.96$ survival)
Bankers' discount factor	β^{int}	0.997	$= 1/R^{bq}$, Bocola Table 1
Steady-state deposit rate	$\bar{r}$	0.003	sample-average risk-free rate
Target leverage	$\bar{\theta}$	4.0	Gertler–Karadi (2011)
Target credit spread	$\bar{\sigma}$	0.005	200 bp p.a. (GK11)
Implied multiplier	$\bar{\mu}$	≈ 0.02	strictly binding
<i>Sovereign debt and risk</i>			
Amortisation rate	δ_b	0.056	Bocola Table 1; duration ≈ 7 y
Debt stock	$\bar{B}$	0.98	bank exposure = 7.6% of assets
Recovery rate	ϱ_D	0.45	Greek PSI 2012
Persistence of s	ρ_s	0.95	Bocola Table 2
Innovation s.d. of s	σ_s	0.63	Bocola Table 2
Rest-point default probability	$\pi^d(\bar{s})$	0.1%	normalisation
<i>Government, trade and policy</i>			
Fiscal elasticity	γ_τ	1.0	Bocola Table 1
Government spending	G	0	normalisation
Home bias	ϖ	0.85	standard
Trade elasticity	η	0.5	standard
External premium	κ_{nfa}	0	frictionless UIP nested
TPI redemption value	ϱ^{tpi}	1.0	near-par floor
TPI real shield	ς	0.5	partial backstop
TPI activation	$\varkappa_{\text{tpi}}$	$\{0, 0.5, 1\}$	per experiment
CB portfolio-balance elasticity	ψ_{cb}	0.5	solver-feasible floor

Table 4: Mapping between the notation of this document and code/global/.

Symbol	Code name	Location
$\varphi_{X,t}$ (franchise value)	alpha_D, alpha_F	point_map.py, bank.py
$\mu_{X,t}$ (IC multiplier)	mu_D, mu_F	point_map.py
λ_X (divertability)	lambda_K, lambda_bD, lambda_bF	calibration.py
$\Omega_{X,t,t+1}$	Om0_D, Om1_D, E_Om_D	point_map.py
$\mathcal{A}_{X,t}$ (bank assets)	assets_D, assets_F	point_map.py
$\lambda\mathcal{A}$ (divertable base)	lev_D, lev_F	point_map.py
$n_{X,t}^g, n_{X,t}$	ng_D, n_D	point_map.py
$P_{X,t}$ (deposit obligation state)	P_D, Pp_D	point_map.py, state_grid.py
$P_{X,t}^c$ (CES price index)	P_CES_D	trade.py
q_t^D, q_t^F	Q_bD, Q_bF	point_map.py
$Q_{X,t}$ (Tobin's q)	Q_D, Q_F	capital.py
h_t (survival factor)	surv, surv_d	point_map.py
$\pi^d(s)$	default_prob(s)	state_grid.py
$r_{X,t}^{wc}$	r_wc_D, wedge_sp_D	point_map.py
$\varkappa_{\text{tpi}}, \varrho^{\text{tpi}}, \varsigma$	tpi_activation, recovery_tpi_D, tpi_real_shield	calibration.py
$\bar{T}_X$ (accounting anchor)	hh_T_D, hh_T_F	recursive_main.py
$\tilde{\beta}_X$	beff_D, beff_F	point_map.py
$\mathcal{S}_t$ (state vector)	STATE_NAMES	state_grid.py
$\mathbf{x}$ (seven unknowns)	SOLVE7	decision_rules.py